

Sample Math Document for Docling Testing

1 Basic Algebra

The quadratic formula is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

2 Calculus

The definition of a derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (2)$$

A definite integral:

$$\int_0^\pi \sin(x) dx = 2 \quad (3)$$

3 Linear Algebra

A matrix and its determinant:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det(A) = ad - bc \quad (4)$$

4 Probability

The normal distribution probability density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad (5)$$

5 Series

Euler's identity and an infinite series:

$$e^{i\pi} + 1 = 0 \quad (6)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad (7)$$

6 Inline Math

Here is some text with an inline formula, $E = mc^2$, embedded directly in a sentence. Another inline example: the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ appears frequently in nature.